

1. Project Overview

Welcome to the NexaTel Customer Churn Analytics project. You have been hired as a Junior Data Analyst at NexaTel Digital Services Pvt. Ltd., a fast-growing Indian telecom and digital-services operator. The company's leadership team needs data-driven answers to one urgent question — why are customers leaving? — and that is where you come in.

This project simulates a real-world analytics assignment. You will work with a realistic dataset covering the company's customer base and operations, containing roughly 757,000 records across 24 interconnected tables. Your job is to clean, explore, analyse, and present the data in a way that helps non-technical managers understand who is churning, why, how much revenue is at risk, and what can be done about it.

1.1 Business Context

NexaTel operates as an integrated telecom provider serving customers across India. It sells across multiple product lines — Prepaid Mobile, Postpaid Mobile, 4G, 5G, Fiber Broadband, Enterprise Connectivity, IoT Solutions, Smart Home, and OTT Bundles. The platform accepts multiple payment methods including UPI, Credit Card, Debit Card, Net Banking, Wallet, Auto Debit, Cash, IMPS, NEFT, and RTGS, and serves metro, Tier-2, and Tier-3 cities.

Over the last four quarters, blended monthly churn has risen from 1.9% to 2.7% while the cost of acquiring a new subscriber has climbed 22%. Management has the following pressing questions they need answered this quarter:

- Which plans and customer segments have the highest churn — and why?
- Are unresolved or repeat complaints driving customers away?
- Which customers are the most valuable, and which of them are most at risk?
- Are there network-quality black spots causing localised churn spikes?
- Which states and circles are losing the most subscribers?
- Do our retention campaigns actually keep customers, or are we wasting spend?

2. Dataset Description

You are provided 24 CSV files that together form the NexaTel database. Together they contain approximately 757,000 records and preserve full referential integrity through primary and foreign keys.

2.1 Files Provided

File	Records	Size	Description
customers.csv	19,076	4.5 MB	Customer profiles, segment, plan, tenure, churn status
subscriptions.csv	20,523	1.2 MB	Active and terminated subscriptions per customer
plans.csv	25	2 KB	Plan catalogue: charges, data/voice quota, billing cycle

plan_history.csv	19,349	1.1 MB	Plan upgrade / downgrade change events
contracts.csv	10,607	798 KB	Contracts for postpaid / fibre / enterprise customers
devices.csv	19,000	1.5 MB	Device per customer (handset / router / IoT)
billing.csv	115,313	8.7 MB	Monthly invoices with base, GST and total amounts
payments.csv	92,362	5.9 MB	Payments made against invoices
recharges.csv	146,501	7.2 MB	Prepaid recharge transactions
usage_voice.csv	57,000	2.2 MB	Monthly voice usage per customer
usage_sms.csv	57,000	1.9 MB	Monthly SMS usage per customer
usage_data.csv	57,000	2.1 MB	Monthly data (GB) usage per customer
network_quality.csv	57,000	3.1 MB	Monthly network experience per customer
support_tickets.csv	31,490	2.6 MB	Support tickets, priority, FCR, resolution time
complaints.csv	16,584	1.2 MB	Complaints with type, severity, resolution status
customer_feedback.csv	28,499	1.6 MB	CSAT / NPS survey responses
retention_campaigns.csv	6,475	524 KB	Customer-level retention offers and outcomes
marketing_campaigns.csv	40	3.8 KB	Campaign definitions, budget, reach, conversions
employees.csv	800	84 KB	Support / sales staff handling tickets
stores.csv	2,180	146 KB	Retail stores by city and region
regions.csv	4	< 1 KB	Operating zones (North / West / South / East)
cities.csv	68	3.7 KB	Cities with tier and PIN prefix
states.csv	36	< 1 KB	Indian states and union territories
data_quality_issue_log.csv	20	< 1 KB	Catalogue of intentionally injected DQ issues

2.2 Key Columns Reference

Below are the most important columns you will use throughout this project. Refer to the full Data Dictionary for every column definition.

customers.csv — The Central Table

Column	Type	Example	Description
customer_id	VARCHAR	C0000001	Unique customer ID (primary key)
customer_segment	VARCHAR	Premium	Mass / Value / Premium / Enterprise
product_line	VARCHAR	Fiber Broadband	Primary product the customer is on
plan_id	VARCHAR	PL015	Links to plans.csv
city_name / city_tier	VARCHAR	Pune / Metro	Location and city tier
acquisition_date	DATE (DD-MM-YYYY)	12-07-2021	When the customer joined
tenure_months	INT	57	Length of relationship in months

annual_income_inr	DECIMAL	850000.00	Estimated annual income in INR
customer_status	VARCHAR	Active	Active / Churned
churn_date	DATE	05-02-2026	Date the customer churned (blank if active)

subscriptions.csv — Plans Held

Column	Type	Example	Description
subscription_id	VARCHAR	S0000001	Unique subscription ID
customer_id	VARCHAR	C0000001	Links to customers.csv
plan_id	VARCHAR	PL009	Links to plans.csv
billing_cycle	VARCHAR	Monthly	Monthly / Annual
monthly_charge_inr	DECIMAL	999.00	Recurring charge in INR
subscription_status	VARCHAR	Active	Active / Terminated

billing.csv — Invoices

Column	Type	Example	Description
invoice_id	VARCHAR	INV00000001	Unique invoice ID
customer_id	VARCHAR	C0000001	Links to customers.csv
billing_date	DATE (DD-MM-YYYY)	01-03-2026	Date the invoice was raised
base_amount_inr	DECIMAL	999.00	Charge before GST
gst_amount_inr	DECIMAL	179.82	18% GST component
total_amount_inr	DECIMAL	1178.82	Total payable in INR
payment_status	VARCHAR	Paid	Paid / Unpaid / Overdue

complaints.csv & support_tickets.csv — Service Experience

Column	Type	Example	Description
complaint_id / ticket_id	VARCHAR	CMP00000001	Unique ID
customer_id	VARCHAR	C0000001	Links to customers.csv
status	VARCHAR	Unresolved	Resolved / Unresolved / Pending
severity / priority	VARCHAR	Major / High	Severity or priority level
first_contact_resolution	VARCHAR	No	Was it solved on first contact (tickets)
resolution_hours	DECIMAL	36.5	Hours taken to resolve (tickets)

usage_data.csv — Monthly Data Usage

Column	Type	Example	Description
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usage_id	VARCHAR	UD000000001	Unique usage record
customer_id	VARCHAR	C00000001	Links to customers.csv
usage_month	TEXT	03-2026	Month of usage (MM-YYYY)
data_gb_used	DECIMAL	42.6	Data consumed in GB
data_sessions	INT	540	Number of data sessions

3. Project Tasks & Deliverables

The project is divided into 4 phases. Complete them in order — each phase builds on the last. The estimated time per phase is shown below.

Phase	Deliverable	Skills Used	Difficulty
Phase 1 (Week 1)	Data Loading, Cleaning & Quality Report	Python / Excel, Pandas	Beginner-Intermediate
Phase 2 (Week 2)	Exploratory Data Analysis (EDA) with Charts	Pandas, Matplotlib / Seaborn	Intermediate
Phase 3 (Week 2-3)	Churn KPI Calculations & Insights	Python / SQL / Excel	Intermediate
Phase 4 (Week 3-4)	Dashboard + Management Summary Report	Power BI / Excel / Python	Intermediate

3.1 Phase 1 — Data Cleaning & Quality Check (Est. 6-7 hours)

Before any analysis, you must understand and clean the data. A real analyst never trusts raw data blindly. Work through these steps:

- Load all 24 CSV files into Python (pandas) or your tool of choice.
- For each table, run a basic profile: row count, column count, and data types.
- Check for missing values in every column — report which columns have nulls and what % of rows are affected.
- Check for duplicate rows — especially duplicate customer_id in customers.csv and duplicate invoice_id in billing.csv.
- Validate data types — dates should parse as dates (note the DD-MM-YYYY format), amounts should be numeric.
- Check for outliers in data_gb_used and negative values in total_amount_inr. Flag them.
- Verify referential integrity — do all customer_id values in subscriptions, billing and usage exist in customers.csv?
- Compare your findings against data_quality_issue_log.csv, then write a 1-page Data Quality Report.

Specific checks to run:

- billing.csv: are there any future billing_date values? Any negative total_amount_inr?
- customers.csv: any impossible ages (negative or > 120)? Any invalid PIN codes or phone numbers?

- payments.csv: does every invoice_id in payments exist in billing.csv? Count the broken ones.
- complaints.csv: what share of complaints are still Unresolved?
- subscriptions.csv: any rows with a null plan_id or an invalid subscription_status?

Deliverable: A Python notebook (or Excel sheet) with your cleaning code and a written summary of findings. Name it: Phase1_DataCleaning_[YourName].ipynb

3.2 Phase 2 — Exploratory Data Analysis (Est. 7-9 hours)

EDA is about understanding patterns, distributions, and relationships. Use charts to tell the churn story. Create the following charts and write 2-3 sentences of observation for each:

- Overall churn rate — a KPI card plus a bar chart of churned vs active customers.
- Churn by billing cycle — compare Monthly vs Annual plans. Which churns more?
- Churn by tenure band — a line or bar chart across 0-3m, 3-12m, 1-3y, 3y+.
- Churn by number of unresolved complaints — does churn rise with complaints?
- Churn by customer segment — Mass / Value / Premium / Enterprise.
- Top 10 states/circles by churned customers — a horizontal bar chart.
- Average data usage by city tier — Metro vs Tier-2 vs Tier-3 (histogram or bar).
- ARPU distribution — a histogram of monthly charge across the base.
- Network quality vs churn — scatter or grouped bar of dropped-call rate vs churn.
- Retention campaign outcomes — accepted / declined / no response, and retained share.

Deliverable: A Python notebook with all 10 charts. Save charts as PNG images too. Name it: Phase2_EDA_[YourName].ipynb

3.3 Phase 3 — Churn KPI Calculations (Est. 7-9 hours)

Now calculate the business metrics management actually asks about. These are the numbers that go into board reports.

KPI / Metric	Formula	Why It Matters
Customer Churn Rate	$\text{Customers Lost} / \text{Customers at Start} \times 100$	Target < 2.1% monthly. The headline retention metric.
Customer Retention Rate	$(\text{End} - \text{New}) / \text{Start} \times 100$	The positive mirror of churn.
ARPU	$\text{Total Recurring Revenue} / \text{Avg Active Subscribers}$	Monetisation per subscriber per month.
Customer Lifetime Value	$\text{ARPU} \times \text{Gross Margin \%} \times (1 / \text{Churn Rate})$	Premium should be far higher than Mass.
Revenue Lost to Churn	$\text{Churned Customers} \times \text{their ARPU}$	Translates churn into rupees at risk.
Monthly Recurring Revenue	Sum of active recurring charges	Size of the predictable revenue base.
First Contact Resolution	$\text{Resolved on First Contact} / \text{Total} \times 100$	Target > 78%. Strong churn lever.
Avg Resolution Time	Mean of (resolved - created) hours	Target < 24h. Long waits push customers out.
Contract Renewal Rate	$\text{Contracts Renewed} / \text{Due for Renewal} \times 100$	Target > 88% for postpaid / fibre.

Avg Customer Tenure	Mean tenure_months of active base	Rising tenure = improving stickiness.
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Intermediate Challenge (Optional, Bonus): Build a cohort retention analysis. Group customers by their acquisition month (cohort) and, for each cohort, compute the share still active after 3, 6, and 12 months. Hint: use pandas pivot_table with acquisition_date and churn_date together.

Deliverable: A notebook (or Excel workbook) with all KPI calculations clearly labelled. Name it: Phase3_KPIs_[YourName].ipynb

3.4 Phase 4 — Dashboard & Final Report (Est. 9-11 hours)

Dashboard Requirements (Power BI or Excel)

Build a dashboard with the following 5 pages / sections:

- Page 1 — Executive Overview: churn rate card, retention rate, revenue-at-risk, monthly churn trend.
- Page 2 — Customer & Segment: churn by segment, tenure bands, ARPU distribution, top at-risk customers.
- Page 3 — Service & Support: FCR, avg resolution time, complaints by status, churn vs unresolved complaints.
- Page 4 — Network & Regional: churn by state/circle, network-quality map, dropped-call vs churn.
- Page 5 — Campaigns & Revenue: retention campaign outcomes, MRR trend, revenue lost to churn.

Each page must have: a title, at least 2 charts, at least 1 filter/slicer (e.g. date range, segment, or region), and a text box with 1-2 key insights.

Management Summary Report

Write a 2-3 page report (in Word or PDF) addressed to the Chief Customer Officer of NexaTel. The report should include:

- Executive Summary (5-6 sentences): What is the overall retention health of the base?
- Top 3 Findings: the most important churn drivers you discovered.
- Risks Identified: which numbers are most concerning?
- Opportunities: where should retention effort be focused to protect the most revenue?
- Recommendations: at least 3 specific, actionable suggestions.

Important: Write in plain language. Avoid jargon. A manager who does not know Python should be able to read and understand your report.

Deliverable: Dashboard file (.pbix or .xlsx) + Management Report (.docx or .pdf). Name them: Phase4_Dashboard_[YourName] and Phase4_Report_[YourName].

4. Evaluation Rubric

Your submission will be evaluated on the following criteria. Total marks: 100.

Criteria	Marks	What We Look For
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Phase 1: Data Cleaning & QC Report	15	Correct null/duplicate/integrity checks, clear documentation, logical conclusions.
Phase 2: EDA Charts (10 charts)	20	Charts clearly labelled, correct chart type, meaningful observations.
Phase 3: KPI Calculations	25	All 10 KPIs correct, formulas sound, values make business sense.
Phase 4: Dashboard Quality	20	Clean dashboard, working filters, accurate data, all 5 pages complete.
Phase 4: Management Report	15	Clear writing, correct figures cited, actionable recommendations, no jargon.
Documentation & Professionalism	5	Tidy notebooks, sensible file names, reproducible work.
Bonus: Cohort Retention Analysis	+5	Correctly grouped by acquisition month, retention pattern is visible.

5. Submission Checklist

Before submitting, verify every item below is complete:

- Phase1_DataCleaning_[YourName].ipynb — Notebook with cleaning code and QC report.
- Phase2_EDA_[YourName].ipynb — Notebook with all 10 charts (charts also saved as PNG).
- Phase3_KPIs_[YourName].ipynb — Notebook or Excel with all 10 KPI calculations.
- Phase4_Dashboard_[YourName].pbix/.xlsx — Dashboard with 5 pages and slicers.
- Phase4_Report_[YourName].docx/.pdf — 2-3 page management summary report.
- All files named correctly with your name.
- No hardcoded numbers in your report — every figure should trace back to your analysis.
- Your report does NOT share any raw data files — only your analysis outputs.

6. Tools & Resources

You may use any of the following tools. Python is recommended for cleaning and KPI work.

- Python (pandas, matplotlib, seaborn) — data cleaning, EDA, KPI calculations.
- Jupyter Notebook — for running and documenting your Python code.
- Power BI Desktop (free) — for the final dashboard.
- Excel (optional) — acceptable alternative to Power BI for the dashboard.
- SQL (optional) — load the CSVs into MySQL / PostgreSQL / SQLite and query them.
- VS Code or PyCharm — recommended Python editors.

Helpful Python libraries to install:

```
pip install pandas matplotlib seaborn openpyxl numpy
```

Hints for Common Challenges

- Date parsing: use `pd.to_datetime(df['acquisition_date'], format='%d-%m-%Y', errors='coerce')`.

- Joining tables: use `pd.merge(subscriptions, customers, on='customer_id', how='left')`.
- Churn rate: `churned = (customers.customer_status == 'Churned');` `rate = churned.mean()*100`.
- Grouping by month: `df['billing_date'].dt.to_period('M').value_counts().sort_index()`.
- INR formatting: use `f'Rs.{value:,.0f}'` for currency display in Python.

7. Important Notes

- Academic integrity: all analysis must be your own. You may discuss approaches with peers, but code and reports must be written independently.
- If you get stuck, describe what you are trying to do in your notebook as a comment — partial work that shows your thinking still earns partial marks.
- You do NOT need to use all 24 tables for every phase. Focus on the most relevant tables for each task.
- The `data_quality_issue_log.csv` lists every intentionally injected issue — use it to check your cleaning.